

# Diagnosis: what the replication's agents said to each other

The 150 two-agent runs of the REPLICATION study — free-text messaging, before any protocol existed. 703 messages over 147 attempts.

## WHY WE LOOKED — the replication left a question open

The replication put two agents on one codebase and found they write working code but cannot combine it: **12% of paired runs delivered both features, against 41% when a single agent did the same work alone**. The pairs could message each other freely the whole time. So the six protocols on slide 2 are not guesses — they are chosen from what those conversations, and the code the agents actually wrote, show about *why* the messaging did not help.

## HOW IT IS DONE — replication\_messages.py over logs/flash\_msg\_\* (the replication's messaging runs)

Each message feature is a **literal string search**: we join an attempt's messages into one string and test for a marker — a filename token such as `loaders.py` for “names a file”, a `?` for “asks a question”, a code fence or `def name(` for “sends code”. When an agent finishes, its work is saved as a **diff**: a machine-readable list of which files it changed and which line numbers it touched — so for every attempt that clashed, we open *both* agents' diffs and look for line ranges that overlap, which pinpoints the collision in the code. Checking those filenames against the conversation establishes the central claim below: the agents had already discussed the file they collided in.

### ① Measured from the message logs

| Measure                             | Value | What the machine literally looks for                                         |
|-------------------------------------|-------|------------------------------------------------------------------------------|
| They name at least one <b>file</b>  | 100%  | text contains <code>x.py / .go / .rs ...</code>                              |
| Either agent asks a <b>question</b> | 9%    | text contains “?”                                                            |
| They exchange <b>actual code</b>    | 3%    | text contains <code>```</code> or “ <code>def name(</code> ” — a lower bound |
| Messages per attempt (median)       | 5     | count of messages in the log                                                 |
| Whole conversation (median)         | 114s  | last timestamp minus first                                                   |

### ② Measured from the code the agents wrote — the evidence the argument rests on

| On the attempts that failed to combine    | Value | How it is computed                                 |
|-------------------------------------------|-------|----------------------------------------------------|
| Clashing attempts examined                | 93    | the combine failed and both agents' work was saved |
| ...both changed the <b>same file</b>      | 100%  | compare the two file lists for a name in both      |
| ...that file was <b>named in the chat</b> | 100%  | search the conversation for that filename          |
| ...they changed the <b>same lines</b>     | 90%   | compare the two line ranges for an overlap         |
| ...both rewrote the <b>same function</b>  | 27%   | compare the function names each one edited         |

Each row compares the two agents' saved changes against each other.

### One attempt, verbatim (it clashed)

|                |                                                                                                                                                                                                         |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>agent 2</b> | <i>“Suggest final combined order: (environment, value, attribute, default=None, separator='.', reverse=False) — just append your reverse param after mine.”</i>                                         |
| <b>agent 1</b> | <i>“my working copy doesn't show your separator changes... we appear to be in separate sandboxes. Since each of us submits our own work independently, this should be fine — no actual clobbering.”</i> |

Agent 2 spells out the correct combined line; agent 1 agrees; **neither types it**. One documented instance of the mechanism.

## They agree on a description of the answer. But the combine is decided by the exact characters they type.

**Scope.** Every figure here is a literal count or a diff comparison, so each one is reproducible from the logs by rerunning the script. Scoring conversations for habits like “both said they'd add theirs last” would need a hand-written phrase list, which belongs to a separate coding study with its own reliability check.

## SLIDE 2

## The six protocols the diagnosis selects

Each answers a specific finding from slide 1, and each forces the pair to agree on more of the final code than the rung below

- Slide 1 diagnosed the replication's failure; each rule set here is built to remove one cause it identified. Same model, tasks, marking and automatic combine throughout — **only the cooperation rules change**, so any difference is caused by the rules.
- They form a ladder: agree on **nothing** at the bottom, on **the exact code** at the top.

|   | Rule set                | They must agree on...                 | What it is trying to fix                                                                                                       | Combined / both work |
|---|-------------------------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|----------------------|
| 1 | <b>control</b>          | Nothing — no messaging at all         | The baseline. If the talking is doing real work, removing it should hurt.                                                      | 13% / 2%             |
| 2 | <b>free-text</b>        | Whatever they choose to say           | The original condition, repeated here so the other five have something to beat.                                                | 21% / 3%             |
| 3 | <b>semi-structured</b>  | A filled-in form about their intent   | Tests whether the problem was just <b>sloppiness</b> : forces every message to name its files and intent, or be rejected.      | 16% / 3%             |
| 4 | <b>plan-handshake</b>   | A split of the files, agreed up front | Fixes the missing hand-out of work: they must <b>agree who owns which file and wait for a reply</b> before typing anything.    | 20% / 10%            |
| 5 | <b>designated-coder</b> | One owner per shared file             | Accepts that a file both need cannot be split, so <b>only one agent writes it</b> ; the other sends a written request instead. | 18% / 58%            |
| 6 | <b>coauthor-overlap</b> | <b>The exact merged code itself</b>   | Goes after the clash directly: they agree the shared code word-for-word, then <b>both type the identical text</b> .            | <b>78% / 69%</b>     |

### Why these six, and why in this order

- Slide 1: clashes happen **inside** a file they already discussed, on the **same line**. Agreeing about *files* is too coarse to prevent them.
- So the result is **predictable before running anything**: rules 1–4 agree only at the level of files or intentions and should fail; only 5–6 reach the code itself.
- Rules 3 and 4 are the useful failures — **3**: they already named their files every time, so imposing structure on the message adds nothing. **4**: a real agreed split made the code more *correct*, and no easier to combine.

**TAKEAWAY** *Each rung takes away one more thing the two agents can disagree about. The top rung takes away all of them.*

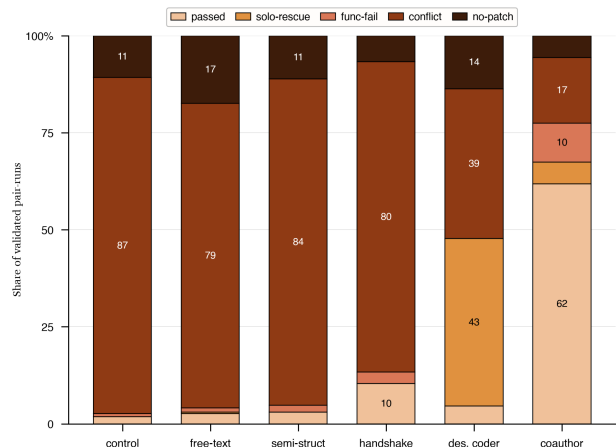
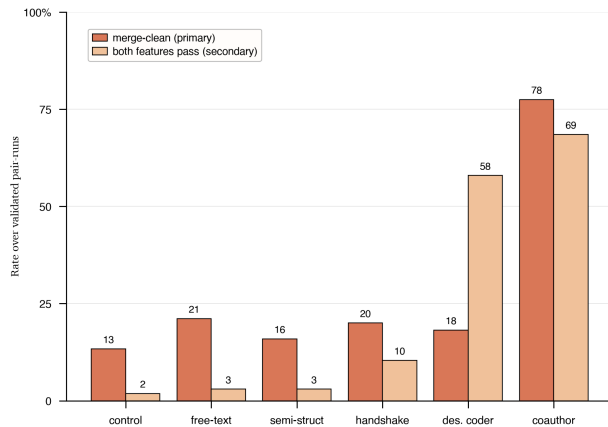
Result column = slide 3's figures, from `analyze.py` over the run logs.

# What happened

18 tasks chosen because the two features genuinely overlap · ~1,250 attempts · Claude Sonnet 5

## READING THE NUMBERS — the two things being measured

|                             |                                                                                                                |
|-----------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Combined cleanly</b>     | The two sets of edits went together automatically. <i>Headline measure.</i>                                    |
| <b>Both features work</b>   | After combining, both features pass their tests. <i>Stricter measure.</i>                                      |
| <b>Clash</b>                | Both rewrote the same lines, so the combine refused.                                                           |
| <b>One carried the pair</b> | Both features work only because <b>one</b> agent's copy already contained both — not because the two combined. |



**How often each rule set worked.** Taller is better. Five sit in the same low band; only coauthor-overlap breaks out.

**How attempts failed.** One bucket each. The clash band dominates everywhere except the last rule set.

| Rule set         | Combined cleanly | Both features work | Clashed |
|------------------|------------------|--------------------|---------|
| control          | 13%              | 2%                 | 87%     |
| free-text        | 21%              | 3%                 | 79%     |
| semi-structured  | 16%              | 3%                 | 84%     |
| plan-handshake   | 20%              | 10%                | 80%     |
| designated-coder | 18%              | 58%                | 39%     |
| coauthor-overlap | 78%              | 69%                | 17%     |

## Key findings

- **Only one rule set worked.** Co-writing the shared code took clean combining **13% → 78%** and both-features-working **2% → 69%** — the only arm to beat the baseline once you account for having tried six things.
- **Talking more achieved nothing.** Tidier messages, or planning who owns which file, scored no better than **no messaging at all** — as slide 1 predicted.
- **It works by changing what gets typed.** The winner produced **26** byte-identical results; the other five produced **1** between them across 1,168 attempts.
- **Working code ≠ combinable code.** designated-coder reaches 58% both-working while clashing as often as the baseline — one agent quietly carried the pair.
- **Not solved.** The best rule set still clashes 17% of the time and breaks another 10%. One model, one setup, one style of automatic combine.

## TAKEAWAY

Rules for cooperation only help when they constrain **what the agents type** — not how much they say to each other.

**Source.** All numbers and both figures from `analyze.py` over `logs/`, frozen in `data/nano_study.json`; charts drawn by `figures.py`. “Beats the baseline” = significant under a Cochran–Mantel–Haenszel test stratified by task, Holm-corrected across the eight comparisons.